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AI for Bharat Hackathon

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Team Name : RootedAI

Team Leader Name : Ganesh K

Problem Statement : Traditional RAG systems only process text documents, 90% of corporate knowledge trapped in multimedia formats like videos, audio recordings, and images inaccessible for intelligent search and retrieval.

Brief about the Idea:

1. This project, called "Adhi," is a production-grade multimodal RAG (Retrieval-Augmented Generation) system that unlocks "dark data" from multimedia sources.

2. It processes videos, audio files, images, and PDFs to extract meaningful content through transcription, OCR, and visual analysis.

3. The system uses asynchronous processing with Celery workers to handle resource-intensive tasks like audio transcription via Sarvam AI and video frame analysis.

4. Users can then query this processed multimedia content using natural language and receive contextual answers with source citations, making previously inaccessible corporate knowledge searchable and actionable.

Your solution should be able to explain the following:

How different is it from existing ideas?

1. Multimodal Processing
2. Asynchronous Architecture(uses Celery and Redis)
3. Smart Task Routing
4. Production-Ready

How will it solve the problem?

1. **Content Extraction**: Converts multimedia to searchable text through transcriptio, OCR, and visual analysis
2. **Intelligent Indexing**: Creates vector embeddings for semantic search across all content types
3. **Unified Query Interface**: Single API endpoint to search across all ingested multimedia content
4. **Source Attribution**: Provides citations with timestamps, page numbers, and visual references

USP of the proposed solution

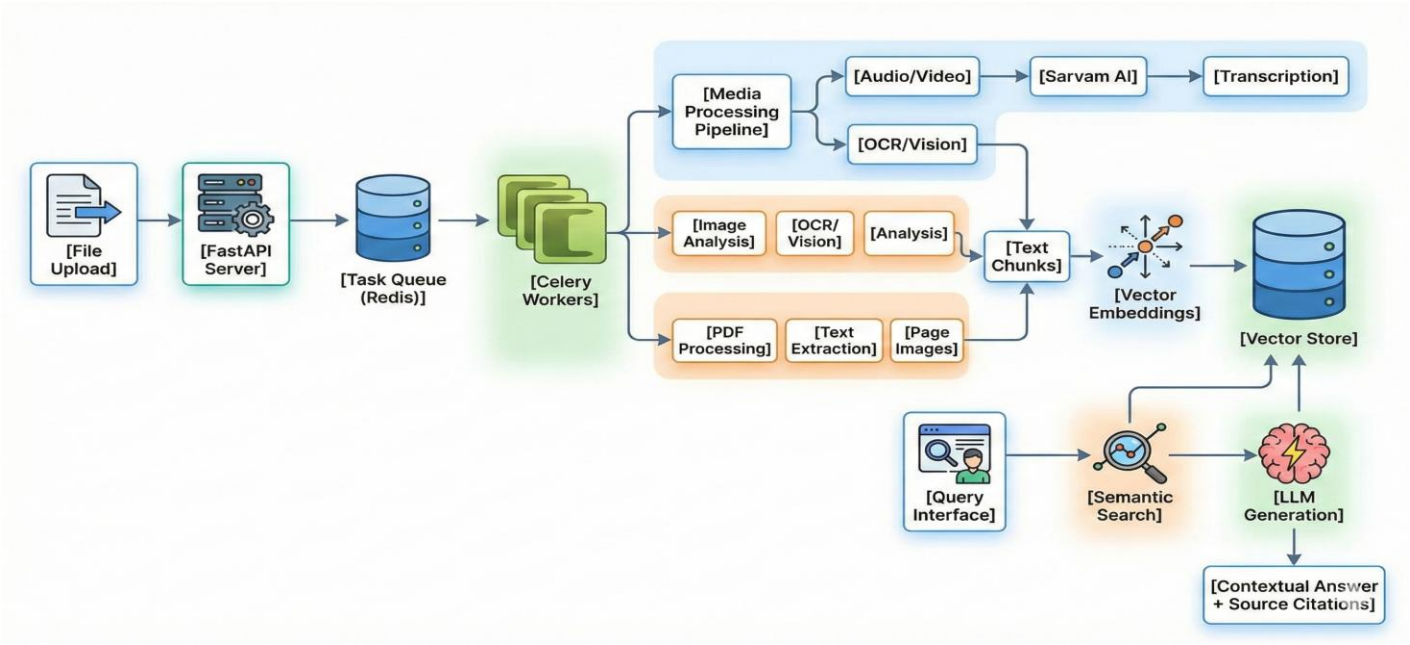
1. **90% Dark Data Unlock**: Specifically targets the untapped multimedia knowledge in organizations
2. **Language Support**: Supports multiple languages including Tamil (ta-IN) through Sarvam AI
3. **Visual Grounding**: Maintains connections between text content and source images/frames
4. **Enterprise-Ready**: Multi-organization support with proper data isolation and scalability

List of features offered by the solution

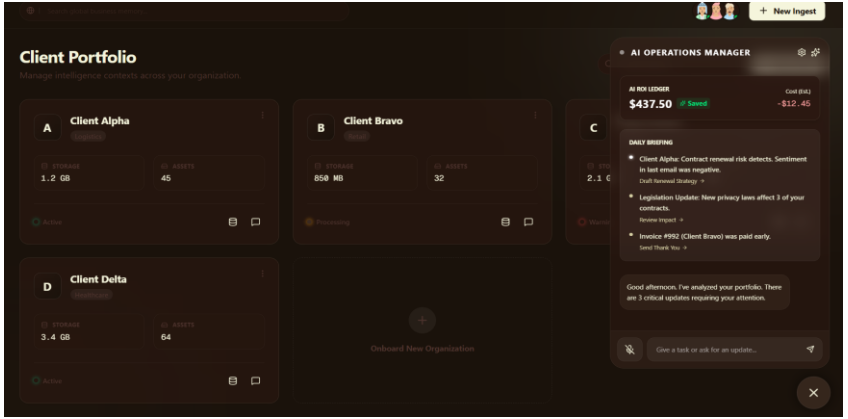
1. **Multi-format Ingestion:** Video (MP4), Audio (various formats), Images (JPG, PNG), PDFs
2. **Asynchronous Processing:** Non-blocking file uploads with background processing
3. **Real-time Status Tracking:** Task progress monitoring with detailed status updates
4. **Intelligent Transcription:** Audio-to-text conversion with language detection
5. **Visual Analysis:** Frame extraction and image content analysis
6. **Semantic Search:** Vector-based similarity search across all content
7. **Source Attribution:** Detailed citations with timestamps and visual references
8. **Multi-organization Support:** Tenant isolation for enterprise deployments
9. **RESTful API:** Complete API with ingestion, query, and status endpoints
10. **Web Interface:** Modern React-based dashboard for file management and querying
11. **Streamlit Frontend:** Alternative interface for quick testing and demos

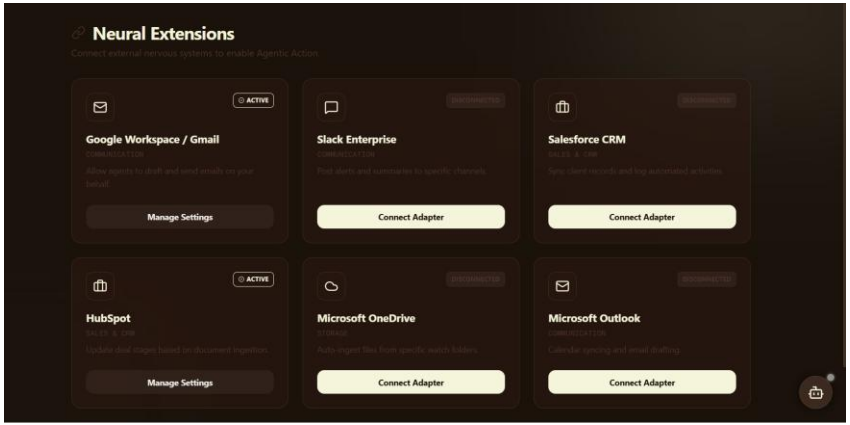
Process flow diagram or Use-case diagram

.

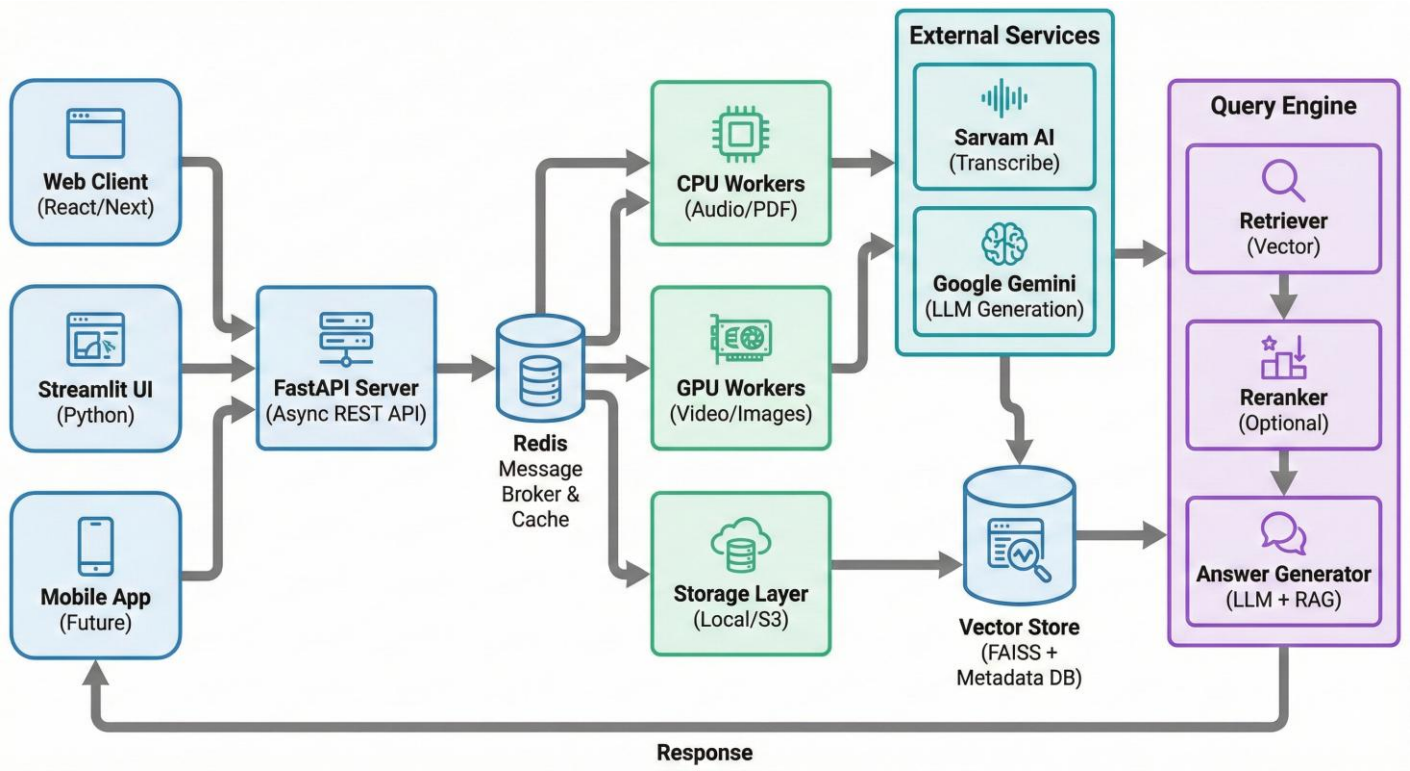


Wireframes/Mock diagrams of the proposed solution (optional)





Architecture diagram of the proposed solution:



Technologies to be used in the solution:

Backend Technologies:

1. Frontend Technologies:
2. Next.js 16: React framework with server-side rendering
3. TypeScript: Type-safe JavaScript
4. Tailwind CSS: Utility-first CSS framework
5. Framer Motion: Animation library
6. React Dropzone: File upload component
7. Lucide React: Icon library
8. Streamlit: Alternative Python-based UI

External Services:

1. Sarvam AI: Audio transcription service
2. Google Gemini: Large language model for answer generation
3. Supabase: Optional cloud database and storage
4. Infrastructure:
5. Docker: Containerization (implied from structure)
6. PostgreSQL: Production database

Technologies to be used in the solution:

Backend Technologies:

1. FastAPI: Async REST API framework
2. Celery: Distributed task queue for background processing
3. Redis: Message broker and caching layer
4. SQLAlchemy: Database ORM with PostgreSQL/SQLite support
5. Pydantic: Data validation and settings management
6. FAISS: Vector similarity search engine
7. Sentence Transformers: Text embedding generation
8. FFmpeg: Video/audio processing
9. OpenCV: Computer vision and image processing
10. Pillow: Image manipulation
11. PyPDF: PDF text extraction
12. pdf2image: PDF to image conversion

Estimated implementation cost (optional):

Category	Line Item	Min Cost (Rupee)
1. Development Phase(One-Time Costs)	Senior Full-Stack Developer (4 months)	₹40,000.00
	ML/AI Engineer (3 months)	₹32,500.00
	DevOps Engineer (2 months)	₹18,700.00
	Total Development Cost	₹91,200.00
2. Infrastructure(Annual Recurring)	Cloud Hosting (AWS/GCP)	₹2,400.00
	Sarvam AI API	₹1,200.00
	Google Gemini API	₹1,800.00
	Storage & Bandwidth	₹1,200.00
	Monitoring & Security	₹600.00
	Total Annual Infrastructure	₹7,200.00
3. Ongoing Ops(Annual Recurring)	Maintenance & Updates (Engineering hours)	₹20,000.00
	Infrastructure (from Section 2)	₹7,200.00
	Total Annual Operating Cost	₹27,200.00
SUMMARY	**TOTAL FIRST YEAR COST**	₹98,400.00

Add as per the requirements for the hackathon:

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Thank You